



Welding Accessories

Projection welding electrodes, adapters, shanks, hex adapters, weld wheels and shunts

Luvata, the world's largest resistance welding electrode manufacturer, also offers a broad range of products and support for your additional welding needs.

Our solutions are proven to result in cost reductions through process optimization. We offer our customers additional benefits including:

- Improved Quality
- Cost Savings
- Just-In-Time Sourcing
- Flexible Packaging
- Excellent Customer Service
- Process and Product Engineering Support

Our welding accessories are offered in a wide range of alloys including:

- Copper Zirconium (Class I)
- Copper Chrome Zirconium (Class II)
- Copper Nickel Silicon (Class III)
- Dispersion Strengthened Copper (Class 20)



About Luvata

Luvata is a world leader in metal solutions manufacturing and related engineering services to industries such as renewable energy, automotive, healthcare, and power generation and distribution. The company's continued success is attributed to its longevity, technological excellence and strategy of building partnerships beyond metals. Employing approximately 1,400 staff in 6 countries, Luvata works in partnership with customers such as ABB, CERN, Siemens and Toyota. Luvata is a group company of Mitsubishi Materials Corporation.

Product Range



Projection welding electrodes



Projection welding electrode holders

Nut and stud welding electrodes

- Nut welding electrode assemblies for manual and auto load applications
- Electrode bases

- Electrode holders
- Stud and nut welding electrodes alloys include Class II, Class III and Class 20 (DSC)



Adapters



Weld wheels

Hex and straight length adapters

- Hex and straight length adapters in all lengths and sizes
- Adapter alloys include Class II, Class III and Class 20 (DSC)

Weld wheels

- Weld wheels up to 20"
- Weld wheel alloys include Class I, Class II and Class III

Shunts

- Designed to customer's requirements

Speciality electrodes

- All materials are designed with regards to application

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luvata.com



Kova®

The high-strength, beryllium-free alloy

Kova is a 'beryllium-free' alloy made of copper, nickel, silicon and chromium (CuNiSiCr). Kova is the perfect replacement for harmful beryllium copper alloys.

Kova is a very high-strength, beryllium-free precipitation hardening alloy with medium-high electrical and thermal conductivity.

Advantages

- No harmful ingredients
- Easy disposal
- Easy to machine and form
- High strength with moderate thermal and electrical conductivity
- Extended use due to hardness

Applications

- Adaptors
- Shanks
- Projection welding electrodes



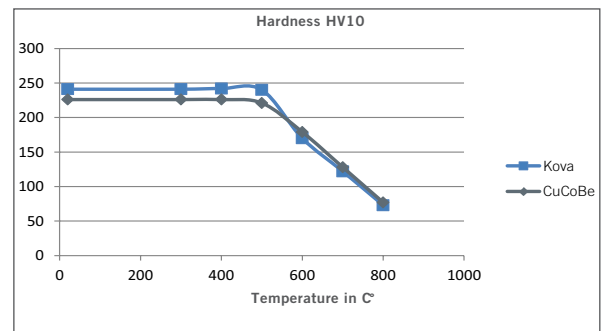
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Specification

Alloy	Kova® NK203 is a RWMA class III alloy		
Chemical composition	Ni 1.8 - 3.0%; Si 0.4% - 0.8%; Cr 0.1 - 0.8% Fe 0.15% max; Cu balance		
Physical, electrical and thermal properties	Density	8.8kg/dm ³	
	Coefficient of linear expansion	0.0000175 1/K	
	Specific heat	380 J/(kg x K)	
	Melting temperature	1020-1040°C	
	Electric conductivity	22-24 MS/m	
	Electrical conductivity (% IACS)	38-42%	
	Electrical resistivity (mass)	0.37-0.41 g/m ³	
	Thermal conductivity (20°C)	220 W/Km	
Documentation	Acceptance test certificate N-CEN/TS 13388:2008		

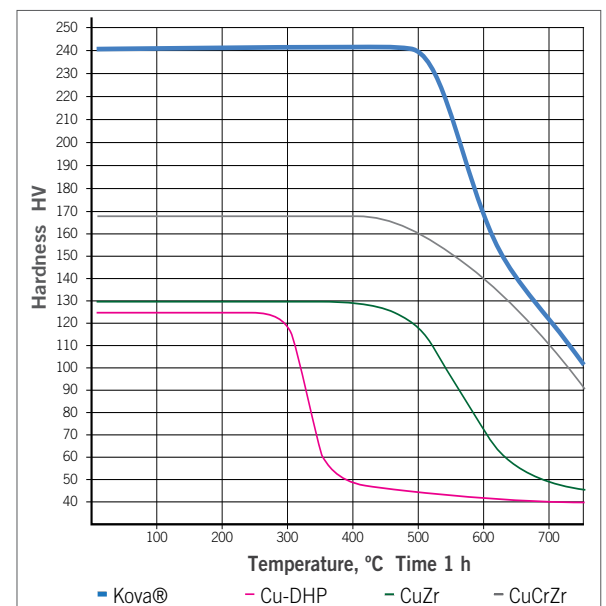
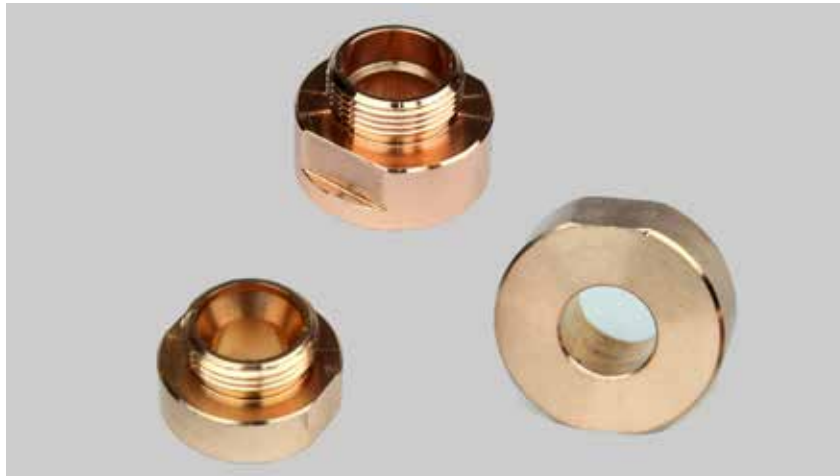


Kova hardness comparison

Mechanical Specifications

Form of supply	Tensile strength [N/mm ²]	0.2% offset yield strength [N/mm ²]	Elongation AS [%]	Hardness HV
Adaptors	650-800	600-750	9-15	220-250

Errors and omissions excepted. Values given are industry standards.



Kova softening behavior - resistance against softening

Room temperature hardness is presented in the figure above as a function of annealing temperature. Material at hard or aged temper.

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